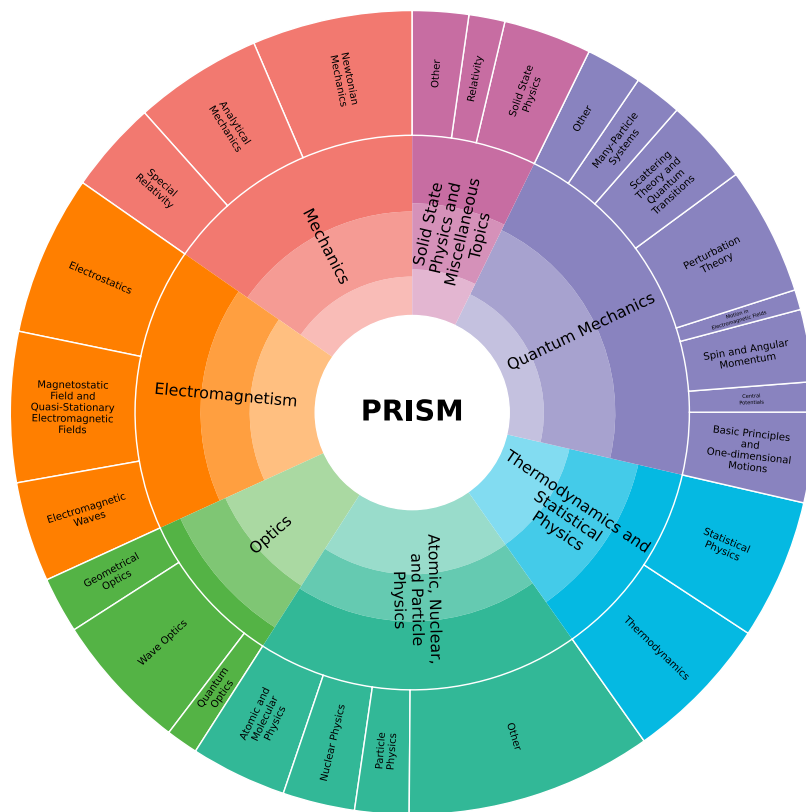


Diagram

Context

Consider a rectangular solid block with length l (along x -direction), cross-sectional area A (the area of face parallel to applied force), shear modulus n (for the block's material), and density ρ . A tangential shearing force F is applied to one face, causing a tangential displacement u at the upper surface and producing a small shear angle φ . For the analysis of elastic waves, let $u(x, t)$ denote the displacement in the y -direction at position x and time t . You should use F for the shearing force, A for the area, n for the shear modulus, l for the block length, u for the tangential displacement, φ for the shear angle, ρ for the density, and υ for the speed of the transverse elastic wave.

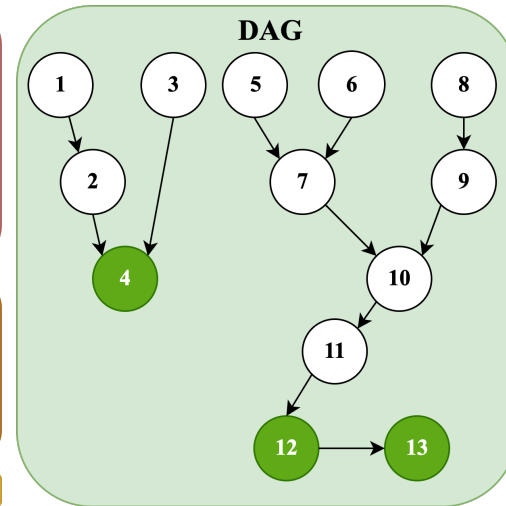
Sub-questions

- A shearing force F is applied tangentially to a rectangular solid block as shown in Fig. Find, within elastic limits, the relation between the tangential displacement u at the upper surface and the applied force F
- The elastic properties of the solid support elastic waves. Assume a transverse plane wave propagates in the x -direction, with oscillations in the y -direction. Derive the equation of motion for the y -direction displacement $u(x, t)$, where u is the displacement at
- Find, in terms of the shear modulus n and density ρ , the speed υ of the transverse elastic wave as described in part (b).

Final answer form: algebraic **Final answer instructions:** Your final answer should be given as a equation to reflect the relationship between... and use only...

Solutions

- Hooke's law for shearing yields $\frac{F}{A} = n \varphi$ (1). The tangential displacement at ... by $u = l \varphi$ (2). But the tangential ... by u , so $u = l \varphi$. Solving for φ $\varphi = \frac{F}{A n}$ (3). Substituting this results into the previous equation gives $u = l \varphi$ $u = \frac{F l}{A n}$ (4).
- The potential ... is $\frac{1}{2} n \varphi^2$ (5). For small ..., $\varphi = \frac{\partial u}{\partial x}$ (6). Thus ... $\frac{1}{2} n \left(\frac{\partial u}{\partial x} \right)^2$ (7). The kinetic energy ... is $\int_0^l \frac{1}{2} \rho A \left(\frac{\partial u}{\partial t} \right)^2 dx$ (8). According to ..., $\Delta \int_{t_1}^{t_2} L dt = 0$, where L is ..., with kinetic energy $S_T = \int_0^l \frac{1}{2} \rho A \left(\frac{\partial u}{\partial t} \right)^2 dx$ (9) and potential energy $S_V = \int_0^l \frac{1}{2} n A \left(\frac{\partial u}{\partial x} \right)^2 dx$ (10). So the action is $S = \int_{t_1}^{t_2} \int_0^l \frac{1}{2} \rho A \left(\frac{\partial u}{\partial t} \right)^2 - \frac{1}{2} n A \left(\frac{\partial u}{\partial x} \right)^2 dx dt$ (11). Applying the Euler-Lagrange equation leads to $\frac{\partial^2 u}{\partial t^2} - \frac{n}{\rho} \frac{\partial^2 u}{\partial x^2} = 0$ (12).
- From wave equation, $\frac{\partial^2 u}{\partial t^2} = \frac{n}{\rho} \frac{\partial^2 u}{\partial x^2}$. The speed of transverse elastic wave is $\upsilon = \sqrt{\frac{n}{\rho}}$ (13).



Difficulty

Medium

Physics Domain

Mechanics

Knowledge Concepts

Analytical Mechanics